AudioBERT: Audio Knowledge Augmented Language Model

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arXiv:2409.08199v2 [cs.CL] 16 Jan 2025

***Abstract*—Recent studies have identified that language models, pretrained on text-only datasets, often lack elementary visual knowledge, *e.g.,* colors of everyday objects. Motivated by this observation, we ask whether a similar shortcoming exists in terms of the *auditory* knowledge. To answer this question, we construct a new dataset called AuditoryBench, which consists of two novel tasks for evaluating auditory knowledge. Based on our analysis using the benchmark, we find that language models also suffer from a severe lack of auditory knowledge. To address this limitation, we propose AudioBERT, a novel method to augment the auditory knowledge of BERT through a retrieval-based approach. First, we detect auditory knowledge spans in prompts to query our retrieval model efficiently. Then, we inject audio knowledge into BERT and switch on low-rank adaptation for effective adaptation when audio knowledge is required. Our experiments demonstrate that AudioBERT is quite effective, achieving superior performance on the AuditoryBench. The dataset and code are available at https://github.com/HJ-Ok/AudioBERT.**

***Index Terms*—Auditory Knowledge, Language Model, Retrieval Augmented Prediction**

I. Introduction

The advance of pretrained language models has spurred significant improvements across various language-related tasks [1], [2], and has been extended to processing multimodal information [3], [4]. However, a major limitation of popular language models is that they are pretrained only on textual data, which can lead to gaps in knowledge from other domains. Indeed, in visual domains, researchers have repeatedly found that common language models lack sufficient visual commonsense knowledge—such as the color of common objects—leading to a poor performance on visual tasks [5]–[8]. In response, recent studies have proposed algorithms to augment language models with visual knowledge [9]–[12].

Does the same limitation hold for *auditory* commonsense knowledge? This question, unfortunately, has not been clearly addressed yet. Although a recent work studies the effectiveness of audio snippet embeddings in the language model representation space [13], it is not known whether the language models have rich commonsense knowledge regarding the auditory signals, e.g., which animal make a specific sound (Fig. 1).

To answer this question, we present **AuditoryBench**, the first benchmark dataset (to our knowledge) for evaluating the language models' auditory knowledge. In particular, we propose two auditory knowledge tasks: (1) animal sound recognition and (2) sound pitch comparison. The animal sound recognition task asks the language model to predict which animal is likely to make a sound that corresponds to specific echoing words (*i.e.*, onomatopoeia), such as "meow." The sound pitch comparison task asks the language model to predict which sound source (*e.g.*, musical instruments, objects, or environments) is likelier to produce sound with a higher pitch. To construct this benchmark, we propose an LLM-based data-processing pipeline for the sake of the scalability of the benchmark dataset [14], [15].

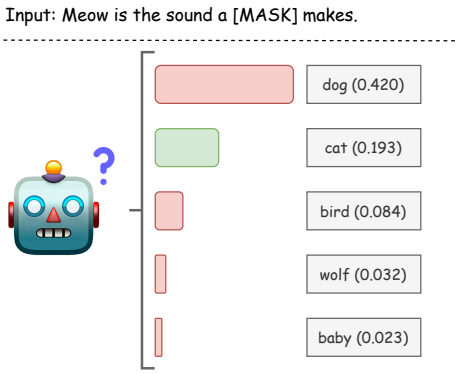

Fig. 1. A visual illustration of the BERT prediction probabilities on a masked word, when asked to complete an example sentence from the AuditoryBench dataset which requires auditory commonsense knowledge.

Using AuditoryBench, we discover that language models severely lack auditory commonsense knowledge. In particular, we test three different language models—BERT [1], Gemma [16], and LLaMA [17]—and find that all models achieve low predictive accuracy in both benchmark tasks (Table II).

To address this shortcoming, we propose **AudioBERT**, a simple yet effective retrieval-based framework for injecting auditory knowledge into language models. Our approach involves detecting text spans where auditory knowledge is necessary. Whenever needed, relevant audio is retrieved by querying the detected text span to CLAP [18], a model that measures the text-audio similarity. Then, the embedding of the retrieved audio sample is injected into the language model. Upon identification of auditory spans by the detector, the language model activates Low-Rank Adaptation (LoRA) [19] weights, which is finetuned with AudiotoryBench, which maintains its pretrained knowledge makes the model perform well in other tasks by deactivating LoRA weights. Our experi-

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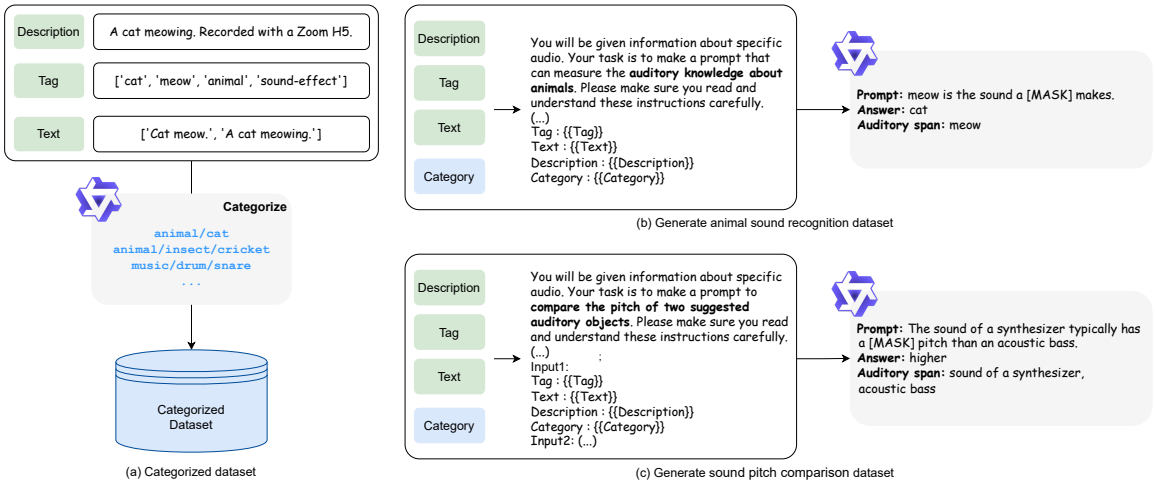

Fig. 2. A visual illustration of the overall data generation process for the AuditoryBench.

ments demonstrate that the proposed method is quite effective, improving the prediction accuracy on AuditoryBench by more than 40% in the test set. As far as we know, AudioBERT is the first algorithm to augment language models with auditory commonsense knowledge.

The key contributions of our work are threefold:

- We introduce the first benchmark dataset for evaluating the auditory knowledge of language models via an efficient LLM-based data generation pipeline.
- We discover that common language models pretrained on text-only datasets suffer from a severe lack of auditory commonsense knowledge.
- We propose a novel and effective framework to inject auditory knowledge into pretrained language models, which adaptively retrieves and utilizes relevant auditory knowledge, utilizing switch-on-off LoRA to preserve pretrained knowledge.

II. Methods

A. AuditoryBench

To assess the auditory commonsense knowledge, we construct a dataset called AuditoryBench. AuditoryBench is based on LAION-Audio-630K [20], a large-scale audio-text dataset with over 600,000 audio-text pairs.

The first step to construct AuditoryBench is to categorize each audio sample of the dataset by processing the paired text information with an LLM (Fig. 2(a)); the text information consists of the audio metadata (description and tag) and the paired text. Based on this information, we give a detailed instruction to categorize hierarchically (*e.g.,* "animal/dog/pomeranian") with few-shot examples.

The next step is to construct tasks based on the generated categories and the text information. We design two tasks: Animal sound recognition and sound pitch comparison.

Animal sound recognition: This task asks to predict an animal from the given onomatopoeia that corresponds to an animal' sound. Each datum consists of a prompt (*e.g.,* "meow is the sound a [MASK] makes"), an answer (*e.g.,* "cat"), and an onomatopoeic span (*e.g.,* "meow"). This triplet is generated by first selecting a datum that is categorized as "animal," and processing its text information with an LLM; see Fig. 2(b) for an overall illustration and the prompt used. To enhance the data quality, we have additionally employed a human annotator to filter our inappropriate animal answers(e.g., "zombie, vampire").

TABLE I
Statistics for Animal Sounds Recognition and Sound Pitch Comparison Data.

Splits	# Captions	# Words/Caption	# Total tokens
Animal sound recognition			
Train	4,211	9.27	39,024
Dev	593	9.39	5,567
Test	1,211	9.30	11,268
Wiki	197	7.01	1,381
Total	6,212	9.21	57,240
Sound pitch comparison			
Train	8,312	18.18	151,081
Dev	1,178	18.12	21,343
Test	2,387	18.19	43,431
Wiki	3,625	12.41	44,987
Total	15,502	16.83	260,842

Using this approach, we have generated a task with 6,015 samples. The dataset is then split into training, development, and test sets with 70%/10%/20% of samples. We have also collected animal sound data from Wikipedia,[1] allocating them as additional test data. This additional set lets us assess the generalizability and quality of the dataset.

Sound pitch comparison: This task asks to compare the pitch of two different sound sources. Each datum consists of a prompt (*e.g.,* "The sound of a synthesizer typically has a [MASK] pitch than an acoustic bass"), an answer (*e.g.,* "higher"), and two auditory spans (*e.g.,* "sound of a

[1]https://en.wikipedia.org/wiki/List_of_animal_sounds

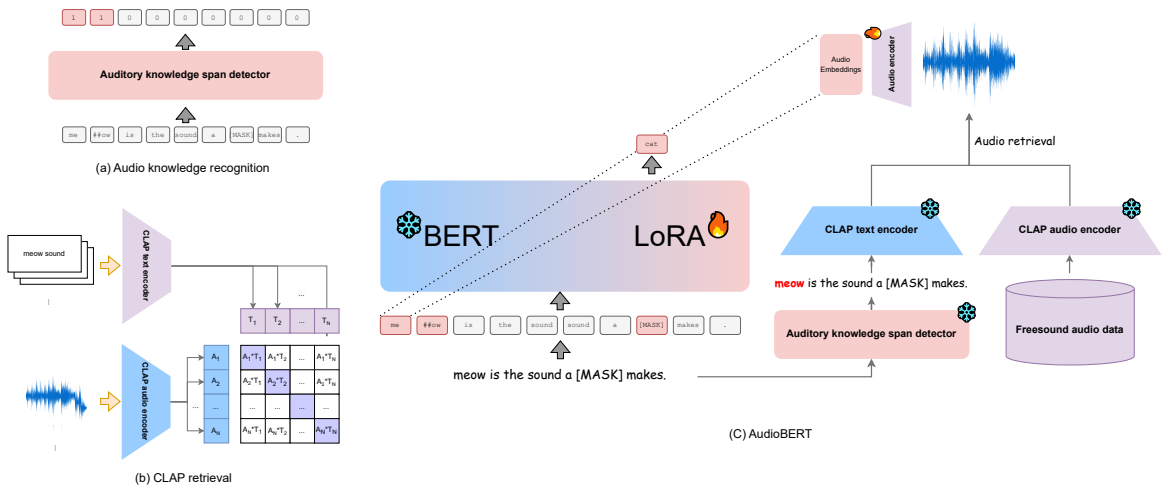

Fig. 3. Overall process of our method

synthesizer," "acoustic bass"). The triplet is generated by first selecting two audio samples, categorized into various music, objects, and environment categories (Fig. 2(c)). Then, using the librosa library [21], we extract the average pitch for each audio sample. We then filter the data by selecting only the pairs in which the pitch difference exceeds 10%, and an LLM can accurately classify which one is higher.

We split the data into training, development, and test sets, with fractions 70%/10%/20%. We have additionally collected the pitch range of musical instruments from Wikipedia[2], to utilize this as a test set. This dataset lets us assess our dataset's generalization and quality.

For both tasks, we use Qwen2-72B-Instruct-AWQ [22] as the LLM. Detailed statistics of these tasks are given in Table I.

B. AudioBERT

Now, we introduce our retrieval-based framework, which is coined AudioBERT. AudioBERT employs two models as its building block: An auditory knowledge span detector, which identifies the spans related to auditory knowledge within the given text, and CLAP [18], a model that is used for retrieval. In what follows, we first describe each component (in Sections II-B1 and II-B2, respectively) and then describe the overall AudioBERT framework in Section II-B3.

1) Auditory knowledge span detector: This model extracts the audio-relevant span from the given text, which can work as a good query. To obtain such model, we simply train a transformer encoder [23] to classify the auditory knowledge span from other tokens, using the cross-entropy loss (Fig. 3(a)).

2) CLAP: Audio-text contrastive learning: To retrieve the relevant audio from the detected auditory knowledge span, we utilize the audio-text model called CLAP [18]. CLAP is trained via audio-text contrastive learning: Given a batch of N audio-text pairs, the cosine similarity between the representations of matched pairs (positive) is maximized, while the cosine similarity is minimized in unmatched (negative) pairs. The loss function is formalized as follows:

$$L_{\text{Audio}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\cos(A_i, T_i)/\tau)}{\sum_{j=1}^{N}\exp(\cos(A_i, T_j)/\tau)} \quad (1)$$

$$L_{\text{Text}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\cos(A_i, T_i)/\tau)}{\sum_{j=1}^{N}\exp(\cos(A_j, T_i)/\tau)} \quad (2)$$

$$L = \frac{1}{2}(L_{\text{Audio}} + L_{\text{Text}}) \quad (3)$$

A_k and T_k are the audio and text embeddings, $\cos(\cdot,\cdot)$ denotes cosine similarity, and τ is a temperature parameter.

3) AudioBERT: AudioBERT injects auditory knowledge into a language model using the auditory knowledge span detector and CLAP. Our approach works in three steps: We retrieve the relevant audio from the database using the span detector and the text encoder of CLAP, generate embedding using the CLAP audio encoder, and add this embedding to the first token of the auditory knowledge span (Fig. 3(c)). During the training, we employ masked language modeling loss and LoRA to finetune the language model's LoRA weights and the audio encoder while keeping the other parameters free. For inference, the auditory knowledge span detector first determines whether auditory knowledge is required. If not, inference proceeds without activating the LoRA weights, maintaining the model's original pretrained weights. However, for tasks that require auditory knowledge, the auditory knowledge span detector recognizes this requirement, retrieves relevant audio, activates the LoRA weights, and injects the audio embedding into the model. This novel architecture enables AudioBERT to dynamically adapt to tasks requiring auditory knowledge while maintaining its performance on general language understanding tasks.

III. Experiments

Evaluation metrics: We employ accuracy and F1-score for our metrics. Higher scores indicate better performance.

[2]https://en.wikipedia.org/wiki/Range_(music)

TABLE II
EXPERIMENT RESULTS OF AUDIO KNOWLEDGE RECOGNITION USING F1-SCORE METRICS.

Trained w/	Animal sound			Sound pitch comparison			Combined		
	Dev.	Test	Wiki Test	Dev.	Test	Wiki Test	Dev.	Test	Wiki Test
Animal sound	92.75 ± 0.37	92.76 ± 0.19	100 ± 0.00	34.87 ± 8.48	34.61 ± 8.17	60.37 ± 12.61	43.67 ± 7.01	43.72 ± 6.77	60.96 ± 12.42
Height of sound	52.63 ± 6.73	52.19 ± 6.40	79.08 ± 13.75	94.76 ± 0.13	95.04 ± 0.09	91.12 ± 2.12	88.64 ± 0.47	95.04 ± 0.43	89.89 ± 2.25
Combined dataset	92.00 ± 0.39	92.67 ± 0.25	100 ± 0.00	94.88 ± 0.24	95.12 ± 0.24	92.47 ± 1.76	94.55 ± 0.19	94.84 ± 0.21	92.06 ± 1.74

TABLE III
EXPERIMENT RESULTS OF AUDIOBERT USING ACCURACY METRICS.

Methods	Animal sound			Sound pitch comparison			Combined		
	Dev.	Test	Wiki Test	Dev.	Test	Wiki Test	Dev.	Test	Wiki Test
BERT-base [1]	15.51	13.46	3.05	59.42	60.41	48.06	44.72	44.61	45.72
BERT-large [1]	16.53	15.85	5.58	59.59	58.90	54.90	45.17	44.41	52.36
RoBERTa-base [24]	14.67	14.04	2.54	54.50	55.84	47.45	41.16	41.77	45.14
RoBERTa-large [24]	16.36	14.70	7.61	55.26	56.64	48.80	42.23	42.52	46.68
Gemma2-2B [16]	14.33	15.11	6.60	59.25	60.45	47.86	44.21	45.19	45.73
LLaMA3.1-8B [17]	23.10	21.80	**16.24**	61.46	62.55	47.72	48.62	48.83	46.10
Ours (BERT-base)	**38.28** ± 0.27	36.63 ± 0.38	14.32 ± 1.66	73.18 ± 0.95	74.83 ± 0.70	55.31 ± 2.38	61.49 ± 0.72	61.97 ± 0.59	53.20 ± 2.34
Ours (BERT-large)	37.30 ± 0.47	**36.69** ± 0.27	15.03 ± 1.17	**75.11** ± 0.80	**76.31** ± 0.60	**56.27** ± 1.18	**62.45** ± 0.69	**62.97** ± 0.49	**54.14** ± 1.18

Implementation details: In our implementation, we employ a BERT-base model for the auditory knowledge span detector. We trained with 5 epochs with a batch size of 16, a learning rate of 1×10^{-5}, and utilizing AdamW [25] optimizer.

For AudioBERT training, we experimented using BERT for the language model and employed an AST [26] encoder for auditory knowledge embedding injecting. We trained with 20 epochs with a batch size of 32, a learning rate of 3×10^{-4}, and utilizing AdamW optimizer. For LoRA, we set the rank and alpha to 64 and 128, respectively.

We report the average score from 5 runs with different random seeds for each setting, utilizing Intel Gaudi 2 and NVIDIA H100.

IV. RESULTS

A. Experiments results

1) Auditory knowledge span detector: We report our auditory knowledge span detector performance in various trained data in Table II. The results suggest that training on single data may perform poorly, but the combined dataset works even with performance gain in sound pitch comparison.

2) AudioBERT: To evaluate the effectiveness of our approach, we compared our model against the existing language models: BERT, RoBERTa, Gemma2-2B, and LlaMA3.1-8B. As shown in Table III, language models show the absence of auditory knowledge, and AudioBERT exhibits competitive performance, demonstrating its effectiveness in augmenting auditory knowledge.

B. Data quality assessment

We took the following steps to ensure the quality and robustness of our data: When constructing the AuditoryBench, we first filtered the data. For animal sound recognition, human annotators filtered out inappropriate labels. For the sound pitch comparison task, we measured the pitch of each audio sample and maintained those correctly labeled. Additionally, we augmented the test set with information from Wikipedia to measure generalization and robustness. Finally, we conducted human evaluations on the test set for each benchmark. Three human annotators evaluated the criteria of accuracy and fluency, scoring out of 100 and 3, respectively. Accuracy reflects whether the labels were correctly applied, while fluency measures grammar, spelling, punctuation, word choice, and sentence structure. As shown in Table IV, AuditoryBench exhibited high quality and robustness.

TABLE IV
THE RESULT OF HUMAN EVALUATION ON THE AUDITORYBENCH DATASET. WE RANDOMLY SAMPLED 100 DATA FROM EACH BENCHMARK AND ASKED THE PARTICIPANTS TO RANK THE OUTPUTS OF DIFFERENT SYSTEMS' ACCURACY AND FLUENCY.

Benchmark	Acc.	Fluency
Animal sound recognition	92.00	2.91
Sound pitch comparison	87.00	3.00

V. CONCLUSIONS

We introduce AuditoryBench, the first benchmark to assess the auditory knowledge of language models and propose AudioBERT, a novel method for injecting auditory knowledge into language models. We believe that our discovery of the lack of auditory knowledge in LMs can have implications for audio-language multimodal research and contribute to the research of language models effectively adapting to diverse modalities.

ACKNOWLEDGMENTS

This work was supported in part by the artificial intelligence industrial convergence cluster development project funded by the Ministry of Science and ICT (MSIT, Korea) & Gwangju Metropolitan City, and in part by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2024-00453301).

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